High-Dimensional Gene Expression Classification Using Regularization and Dimension Reduction

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Description of Problem

Primary Objective: Prediction(Classification) – Build and compare statistical machine learning models to predict cancer type using high-dimensional gene expression measurements from the NCI60 dataset, where the number of predictors greatly exceeds the number of observations *(p >> n)*.

Secondary Analysis:

- Compare performance of multiple statistical learning methods (regularized logistic regression, PCA-based models, SVM, etc.) using cross-validation.
- Study the bias-variance tradeoff by observing how model flexibility/regularization affects training vs validation error.
- Identify a small set of informative genes/features using Lasso/Elastic Net and analyze model interpretability.
- Evaluate the effect of dimension reduction (PCA/PCR/PLS) on predictive accuracy and stability.

Description of the Data

Source: NCI60 gene expression dataset

Dataset Details: The dataset consists of gene expression values for 64 cancer cell lines measured across 6,830 gene. Each observation belongs to a cancer type/class label (multi-class outcome).

Key Variables:

- Predictors (X) : Gene expression value for 6,830 genes (continuous)
- Response(Y): Cancer type label (categorical / multi-class)

Type of Learning

Supervised Learning (Classification):

- Primary task: Predict cancer type from gene expression
- Candidate Models:
  - Baseline: Logistic regression / LDA
  - Regularization: Ridge, Lasso, Elastic Net logistic regression
  - Dimension Reduction: PCA + logistic regression(PCR), possibly PLS
  - Nonlinear: SVM(linear/RBF), Random Forest(depending on results)
- Model selection and evaluation via cross-validation

Data/Statistical Challenges:

- High dimensionality (p >> n) can cause overfitting and non-identifiability in unregularized models.
- Mulit-class classification with small sample sizes may produce high variance estimates, careful cross-validation is required

Technical Considerations:

- Need efficient computation for regularization paths and repeated CV.
- Careful reporting of uncertainty and model stability due to small dataset size.
- Interpretability: selected genes may vary across folds; stability analysis will be included.